DBMS Assignment - 3

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SQL and Query concepts :-

1: What is SQL? Why is it important in DBMS.
Ans→ SQL (Structured Query Language) is a standard programming language used to create, manage and manipulate data stored in relational database management systems (DBMS). It allows users to perform operations such as inserting, updating, deleting and retrieving data from database.

SQL is important in DBMS because:
- It provides an efficient way to interact with database.
- It helps in storing and managing large amounts of data systematically.
- It allows quick retrieval of data using queries.
- It ensures data integrity and security through constraints and permissions.
- It is widely used and supported by almost all relational database systems.

2: What is the difference between DDL and DML?
Ans→ DDL (Data Definition Language) and DML (Data Manipulation Language) are two types of SQL commands used in a database.

- DDL (Data Definition language): DDL is used to define and modify the structure of the database, such as tables and Schemas. Example: CREATE, ALTER, DROP, TRUNCATE.
- DML (Data manipulation language): DML is used to manipulation the data stored in the database tables. Examples: INSERT, UPDATE, DELETE, SELECT.

3. What is a join? Explain its types.

Ans→ A join in SQL is used to combine data from two or more tables based on a related column between them. It helps retrieve meaningful information by connecting tables.

Types of JOINS :-

1. Inner Join: Return only the records that have matching values in both tables.
2. Left join (left outer join): Return all records from the left table and the matched records from the right tables. If no match is found, NULL values are returned for the right table.
3. Right join (Right outer join): Returns all records from the right table and the matched records from the left table. If no match is found, NULL values are returned for the ~~m~~ left table.
4. Full join (full outer join): Returns all records from both tables. If there is no match, NULL values are returned for the missing side.

4. What is GROUP By and HAVING clause?

Date / /
Pg No.

Ans→ The GROUP By and HAVING clause are used in SQL to work with grouped data.

- GROUP BY: It is used to group rows that have the same values in specified columns into summary rows.
  It is often used with aggregate functions like COUNT, SUM, AVG, MAX and MIN.
- HAVING: It is used to filter the grouped data after the GROUP BY operation. It is similar to the WHERE clause but is applied to groups instead of individual rows.

5. What is a Subquery?

Ans→ A Subquery is a query that is nested inside another SQL query. It is used to perform complex operations by first executing the inner query and then using its result in the outer query.
Subqueries are commonly used in SELECT, INSERT, UPDATE or DELETE statements and can helps in filtering or retrieving data based on specific conditions.

6. What is relational algebra?

Ans→ Relational algebra is a procedural query language used in database management systems to retrieve and manipulate data from relational database. It Consists of a set of operations that take one or two relations (tables) as input and produce a new relation as output.
Common operations in relational algebra include :

DELTA
Date / /
Pg No.

- Selection (σ) :- Select rows based on a condition.
- Projection (π) :- Select specific columns.
- Join ($\bowtie$) :- Combines two tables based on a related attributes.
- Set Difference ($-$) :- Returns tuples present in one relation but not in another.

Q: What is the difference between WHERE and HAVING.

Ans⇒ The WHERE and HAVING clauses are both used for filtering in SQL, but they work at different stages.

- WHERE clause :- It is used to filter rows before any grouping is done. It works on individual rows and cannot be used with aggregate function.
- HAVING clause :- It is used to filter data after the GROUP BY operation. It works on grouped data and can be used with aggregate functions.

Design, write and Defend (Query based)

Q: Write an SQL query to retrieve all students with marks > 75.

Ans→

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SELECT *
FROM Students
WHERE marks > 75;
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Explanation :- This query selects all records from the students table where the marks are greater than 75 using the WHERE clause.

9. Write a query using JOIN to combine students and course tables.

Ans → SELECT students.student.name, course.course-name
FROM students
INNER JOIN course
ON student.course-id = course.course-id;
Explanation :- This query uses an INNER JOIN to combine the students and course tables based on the common column course-id. It retrieves the student names along with their corresponding course names.

10. Write a query using GROUP BY to count students per department.

Ans → SELECT department, COUNT (*) AS total_students
FROM students
GROUP BY department;
Explanation → This query groups the records by department and uses the COUNT() function to calculate the number of students in each department.

Case - Based Analytical Questions

11. Write a query to find employees in a specific department.

Ans → SELECT e.employee_name, d.department_name
FROM Employee e
INNER JOIN Department d
ON e.department_id = d.department_id
WHERE d.department_name = 'Sales';
Explanation :- This query uses an INNER JOIN to combine the Employee and Department tables based

on department id. The WHERE clause filters the results to show only employee who belong to the sales department.

12. An e-commerce system stores orders. Retrieve total sales per customer.

Ans→ SELECT customer_id, SUM (order_amount) As total sales
FROM orders
GROUP BY customer_id;

Explanation :- This query groups the data by customer_id and uses the Sum () function to calculate the total sales for each customer.

13. A university database needs top 3 students. Write a query.

Ans→ SELECT *
FROM students
ORDER BY marks DESC
LIMIT 3;

Explanation :- This query [illegible] the students by marks in descending order using ORDER BY and retrieves the top 3 students using the LIMIT clause.

One Query + One Insight

14. Write a SQL query for any real life System and explain logic.

Ans→ SELECT name, salary
FROM employees

Date / /
Pg No.

WHERE Salary > 50000;
Explanation :- This query retrieves the names and salaries of employees whose salary is greater than 50,000. The WHERE clause filter only those records that meet the condition. This can be used in a company system to find high earning employees.

15. Write a JOIN query and mention one use case.
Ans→ SELECT o.order_id, c.customer_name
FROM orders o
INNER JOIN Customers c
ON o.customer_id = c.customer_id;
Explanation :- This query joins the orders and Customers tables using customer_id to display order IDs along with customer names.
Use Case :- used in an e-commerce system to display which customer placed each other.

16. Write a GROUP BY query and explain its output
Ans→ SELECT department, COUNT(*) As total_employees
FROM employees
GROUP BY department;
Explanation :- This query groups employees by department and counts how many employees are in each department. The output will show each department along with the total number of employees in it.